

Question Number	Answer	Additional Guidance	Mark
3(a)(i)	Biuret	Accept copper sulfate and sodium hydroxide (sodium potassium tartrate)	(1)

Question Number	Answer	Additional Guidance	Mark
3(a)(ii)	Blue to {mauve / lilac / purple}		(1)

Question Number	Answer	Additional Guidance	Mark
3(a)(iii)	>2 but <10	ACCEPT between 2 and 10 a range with any values from >2 to <10	(1)

Question Number	Answer	Additional Guidance	Mark
3(a)(iv)	An answer that includes the following points: • use {more concentrations / smaller intervals} (for standards) (1) • between 2 and 10 (1) • use of colorimeter (1)	Not larger range Accept 3-9 in any combination (3-10, 2-9, 3-9)	(2)

Question Number	Answer	Additional Guidance			Mark																	
3(b)(i)	A table with the following features: - suitable table (1) - headings with units correct (1) - all data correctly entered (1)	<table><tr><th rowspan="2">Heating temperature / °C</th><th colspan="2">Protein content (of the animal feed) as percentage of starting content</th></tr><tr><th>(After) 5 days</th><th>(After) 7 days</th></tr><tr><td>7</td><td>67</td><td>28</td></tr><tr><td>17</td><td>53</td><td>26</td></tr><tr><td>27</td><td>38</td><td>24</td></tr><tr><td>37</td><td>25</td><td>22</td></tr></table>			Heating temperature / °C	Protein content (of the animal feed) as percentage of starting content		(After) 5 days	(After) 7 days	7	67	28	17	53	26	27	38	24	37	25	22	(3)
						Heating temperature / °C	Protein content (of the animal feed) as percentage of starting content															
					(After) 5 days		(After) 7 days															
					7	67	28															
					17	53	26															
					27	38	24															
					37	25	22															

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3(b)(ii)	A calculation showing the following steps : - two correct figures from graph, subtracted correctly (1) - divided by temperature difference between the two readings (1) - correct units, percentage (protein reduction) °C⁻¹	accept correct calculation and units on 5 days for up to 2 marks e.g. 28 and 22, 22-28 = (-)6 e.g. (-)6 ÷ (37-7) = (-)0.2 / (-)1/5 Accept per degree C / per °C, / °C Calculation for 5 days gives 1.4 67-25 = (-)42 (-)42 ÷ (37-7) = (-) 1.4	(3)

Question Number	Answer	Additional Guidance	Mark
3(b)(iii)	An answer that includes three of the following points: • loss of protein as temperature rises (after both 5 and 7 days) (1) • heating for {7 days / longer time} reduces protein content more (at each temperature) than does heating for {5 days / shorter time} (1) • {rate of loss ($^{\circ}\text{C}^{-1}$) / gradient} is greater after 5 days (than after 7 days) (1) • higher temperature and longer time have same effect / quantity of protein left at 37 $^{\circ}\text{C}$ {same / similar} (for both times) (1)	ACCEPT negative correlation ACCEPT reverse argument ACCEPT reverse argument	(3)

Question Number	Answer	Additional Guidance	Mark
3(b)(iv)	An answer that includes 6 of the following points: • make (nutrient) agar {plate / broth} with bacterium (1) • use of (safe) named bacterium / do not use pathogenic (1) • description of how to look for the effect of acid (1) • use of water / range of pH (1) • (both) incubated at {same / suitable/ stated} temperature (1) • (both) incubated for {same / suitable / stated} time (1) • method of assessing bacterial growth (1) • use of an example of aseptic technique (1)	e.g acid {on filter paper / in well / added to broth} >10 < 30 1 - 7 days e.g. measure {zone of inhibition / turbidity} e.g. flame loop etc. / disinfect / lit Bunsen / partial lid lifting gloves, goggles in context of acid ref to safe temperature of incubation / ref to safe temperature	(6)